AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 3

The AI Satellite and Constellation Operations Playbook

How AI watches, plans and commands a fleet of dozens or hundreds of spacecraft - without ever mistaking a green dashboard for a healthy fleet.



Gates CO0-CO5 | Eight templates | 20 golden rules

A green dashboard is not a healthy fleet. Operations AI is licensed per spacecraft, per orbital plane, per pass and per failure mode - never in general, and never on an average.

One spacecraft is a mission; eighty-four are a system that fails in chorus

Volume 1 set the licensing rule and Volume 2 designed the envelope. This volume lives inside it, shift after shift. A constellation changes the operations problem in kind, not just in size: passes overlap, anomalies correlate, a software fault arrives on forty spacecraft at once, and the scarcest resource is no longer the satellite - it is operator attention.

AI is genuinely strong here. It can watch every telemetry point on every vehicle, rank a thousand alerts into the five that matter, and draft pass plans faster than any shift team. It is also genuinely dangerous here, for the same reason: at fleet scale, a wrong model does not make one bad call, it makes a coordinated one.

The brief in one page

- **Grade the fleet by its worst spacecraft,** not its average. Averages hide the one vehicle that is dying.
- **Attention is the licensed resource.** Every automation is judged on whether it returns attention to operators or quietly consumes it in supervision.
- **Correlated failure is the design case.** Shared software, shared models and shared ground systems mean the fleet can fail together; plan for chorus, not solo.
- **Nothing commands the fleet that cannot be halted per spacecraft.** A fleet-wide action with no per-vehicle brake is refused, whatever its record.

The operations rule: **AI may watch the whole fleet; it may act only inside a licence written per spacecraft, per plane and per failure mode.**

EXECUTIVE LENS

Fund attention, not dashboards.

MISSION LENS

Ask how the fleet fails together, not one by one.

OPERATIONS LENS

An alert nobody can act on is noise with a salary.

Ten stages, graded stage by stage, with no stage carried by its neighbours

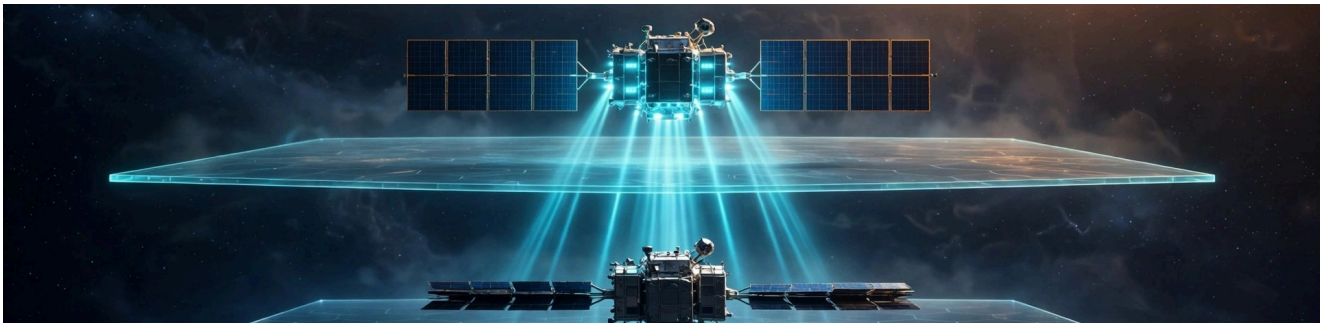


Figure 1 - One spacecraft in three states across stacked time planes: stowed, nominal and safe mode. The fleet is graded by its worst state, never its average.

Constellation operations is a loop, not a line: watch, judge, plan, command, learn. The map below grades each stage for AI licence at CO4, the highest working level short of scale. A stage graded low is not a failure; it is a boundary, and boundaries are what keep a fleet alive.

Stage	What AI does	Licence
1. Telemetry ingest	Stream, time-align and quality-flag every point	CO4 - full
2. Health triage	Rank anomalies by consequence and time-to-harm	CO4 - full
3. Pass planning	Schedule contacts, resolve station conflicts	CO4 - full
4. Command generation	Draft loads inside signed templates	CO4 - checked
5. Command release	Send to spacecraft	Human-held above template class
6. Orbit control	Station-keeping and phasing proposals	CO4 - proposed, human-flown
7. Conjunction response	Screen, rank, draft manoeuvres	Human-decided (Volume 1)
8. Payload tasking	Build collection and service schedules	CO4 - inside signed fences
9. Fleet learning	Mine fleet history for patterns	CO4 - advisory only
10. Disposal watch	Track passivation and deorbit readiness	Human-held, no shortcuts

The pattern is deliberate: the closer a stage gets to irreversible action on orbit, the more human its licence becomes. Ingest and triage can be fully automated because they are reversible and auditable. Release, conjunction execution and disposal stay human because they are not.

EXECUTIVE LENS

The map is the budget: fund stages by licence, not by demo.

MISSION LENS

A low grade is a boundary doing its job.

OPERATIONS LENS

Watch what the loop does at 3 a.m., not at the review.

The first job of operations AI is to decide what deserves a human

An 84-spacecraft fleet can emit millions of telemetry points a day. No team reads that; every team reads a ranking. Triage AI therefore holds real power: whatever it ranks low is, in practice, unseen. That power is licensed accordingly.

How triage earns its licence

- **Rank by time-to-harm,** not by deviation size. A slow battery fade with six orbits to harm outranks a dramatic but harmless mode flap.
- **Show the evidence,** not just the score: the points, the trend and the closest historical analogue travel with every alert.
- **Grade on seeded faults.** Known anomalies are injected into live-like streams monthly; the miss rate, not the demo, sets the licence.
- **Silence is a finding.** A spacecraft that stops reporting oddities it used to report is itself an anomaly; triage watches for missing signals as well as bad ones.
- **Per-vehicle baselines.** Spacecraft age differently. A fleet-wide threshold either cries wolf on the old vehicles or sleeps through the young ones failing.

Triage rule: **an AI that ranks alerts owns what it buries - every suppressed alert is logged, sampled and audited by a human each week.**

EXECUTIVE LENS

Triage quality is a safety metric, report it like one.

MISSION LENS

Judge the ranker on its worst miss, not its average precision.

OPERATIONS LENS

Sample the suppressed pile weekly; that is where surprises live.

Every contact is a negotiation the fleet wins or loses ninety minutes at a time

Low Earth orbit gives each spacecraft a handful of short contacts a day, shared across the fleet and often across rented ground networks. Pass planning is where constellation economics are quietly made: a missed pass is lost data, a late command and sometimes a lost vehicle.

- **Plan the fleet, not the pass.** The optimiser maximises verified mission value across all vehicles - data returned, commands delivered, anomalies checked - never raw contact count.
- **Protect the sick first.** Scheduling priority follows triage: a degrading spacecraft pre-empts a healthy one's routine dump, automatically.
- **Conflicts are surfaced, not silently resolved.** When two high-value needs collide, the planner shows the trade and the price of each choice; silent arbitration hides policy inside software.
- **Weather, station faults and licensing windows** are first-class inputs, re-planned in minutes, with the re-plan logged against the original.
- **Reserved contingency capacity** is never sold to routine traffic: a fixed share of contacts stays empty so the worst day has somewhere to land.

The licence test is a bad day replayed: station outage plus two anomalies plus a conjunction, planned inside the shift. A planner that only shines on quiet days is a fair-weather licence and is graded as one.

EXECUTIVE LENS

Pass efficiency is revenue; pass resilience is survival.

MISSION LENS

Ask what the planner sacrifices when everything collides.

OPERATIONS LENS

Hold contingency contacts empty on purpose, and defend the emptiness.

Nothing reaches a spacecraft that a dumber, independent system has not approved

Volume 1 set the principle: the command checker is deliberately simpler than the planner, built by a different team, on different logic, with one job - to refuse anything outside the signed envelope. At fleet scale that principle is the difference between a bad plan and a bad month.

- **Template classes, not free text.** AI drafts commands only inside pre-signed templates with bounded parameters. Anything outside the template library is a human-authored load, full stop.
- **The checker validates physics and state:** power, thermal, momentum, memory and mode, against the spacecraft's actual current state - not the state the plan assumed.
- **Per-spacecraft brakes.** Any single vehicle can be removed from automated commanding in one action, by any operator, without removing the rest of the fleet.
- **Loads are rehearsed** on the vehicle's digital twin at its current configuration, and the rehearsal result travels with the release request.
- **Every command is attributable:** model version, template, checker verdict and releasing human, retained for the life of the fleet.

Commanding rule: **the planner may be clever; the checker must be right. When they disagree, the spacecraft hears neither until a human decides.**

EXECUTIVE LENS

The checker is cheap insurance against correlated error.

MISSION LENS

A brake that needs a meeting is not a brake.

OPERATIONS LENS

Attribution is what makes automation auditable after the fact.

The fleet's worst hour is rehearsed, graded and owned before it happens

Fault detection, isolation and recovery - FDIR - is where operations AI meets the failure modes Volume 2 designed for. The division of labour is strict: onboard fault protection acts in seconds without asking anyone; ground AI diagnoses in minutes and proposes; humans decide anything that changes the mission.

- **Onboard autonomy stays dumb and fast.** Safe mode entries, load sheds and survival heaters are deterministic, tested logic - never learned behaviour in flight.
- **Ground AI clusters before it concludes.** Simultaneous anomalies across vehicles are treated as one event with one cause until proven otherwise: a ground fault, a bad upload, a space weather event.
- **Recovery plans are drafted, not executed,** with each step's risk and reversibility stated, and the plan rehearsed on the twin before the first step flies.
- **The anomaly library is the fleet's memory:** every event, its signature, its true cause and its fix, searchable by the next shift at 3 a.m. - the test of whether it was written well.
- **Repeat anomalies escalate automatically.** The third occurrence of a signature is a design review, not a procedure.

Grading is by rehearsal: seeded fleet anomalies in the simulator, timed, with the real shift. Response quality on the drill sets the licence for response authority on the day.

EXECUTIVE LENS

Drills are the licence; the real event is just the audit.

MISSION LENS

One event, one cause, until proven otherwise.

OPERATIONS LENS

A library nobody can search at 3 a.m. is an archive, not a memory.

Holding the pattern is a promise made to every other operator in the shell

A constellation is a pattern: planes, phasing and slots that customers, regulators and other operators rely on. AI orbit control keeps that pattern tight at a fraction of the propellant and labour of manual flying - and its errors are also patterned, arriving everywhere at once.

- **Proposals, flown by humans at CO4.** The optimiser drafts station-keeping and phasing manoeuvres per vehicle; a flight dynamics officer releases them in batches after independent propagation.
- **Propellant is fleet life.** Every plan is priced in delta-v and end-of-life date, per vehicle and for the pattern; a plan that saves labour by spending the oldest vehicle's last year is shown as exactly that.
- **Drag and weather are modelled per vehicle,** because sister spacecraft in the same plane decay at measurably different rates.
- **Slot discipline is a safety property:** drifting vehicles are flagged early, while correction is cheap, not when they become someone else's conjunction.
- **End-of-life is planned from day one:** disposal delta-v is fenced in the budget and never traded away by the optimiser.

Orbit rule: **the pattern is a promise. An optimiser that keeps it by spending disposal reserve or the weakest vehicle's life has broken the promise to keep the picture pretty.**

EXECUTIVE LENS

Price every plan in years of fleet life.

MISSION LENS

Independent propagation is the release fee.

OPERATIONS LENS

Disposal delta-v is fenced money; treat raids on it as theft.

Eighty-four spacecraft means the screening never stops - and neither does the deciding

Volume 1 set the standing rule: AI screens, ranks and drafts; humans decide manoeuvres. At constellation scale that rule is tested by volume - hundreds of screening events a week, most routine, a few lethal, all arriving on the same shift.

- **Triage by consequence and confidence:** probability, miss distance, object type and tracking quality combined; low-quality data escalates rather than averages away.
- **Routine events follow a signed playbook** with pre-approved manoeuvre families, so human decision time is spent where judgement actually differs.
- **Coordination is part of the product:** screening data shared with other operators, manoeuvre intent communicated early; a fleet that manoeuvres silently exports its risk.
- **Post-manoeuvre verification is automatic:** the fleet re-screens after every avoidance burn and proves the new geometry before the event is closed.
- **Workload is a safety metric:** when event volume exceeds the decided human capacity, the answer is more capacity or fewer spacecraft per operator - never a quieter threshold.

The refusal from Volume 1 stands unchanged at fleet scale: autonomous conjunction execution is not licensed. Scale is an argument for better screening, not for removing the decider.

EXECUTIVE LENS

Volume tests the rule; it does not repeal it.

MISSION LENS

Silent manoeuvres export risk to neighbours.

OPERATIONS LENS

If the queue outgrows the deciders, grow the deciders.

The payload earns the money; the bus keeps the promise - tasking serves both, in that order of safety

Whether the payload images the Earth, carries traffic or senses the environment, tasking is where customer promises meet orbital reality: power, downlink, weather, exclusion zones and the health of the vehicle itself.

- **Tasking lives inside signed fences:** per-customer priorities, per-region restrictions, per-vehicle health limits, all stated as explicit constraints the scheduler cannot trade away.
- **Health outranks revenue automatically.** A degrading vehicle sheds payload load by rule, not by an operator remembering to ask.
- **Every rejection is explainable** to the customer in plain terms: which constraint bound, and when the next opportunity opens.
- **Quality is verified, not assumed:** delivered products are sampled against the tasking promise - coverage, latency, resolution - and misses feed scheduling, not just apologies.
- **Restricted tasking is refused by design:** exclusion lists and licensing conditions are enforced in the scheduler, audited from its logs, not from its intentions.

Tasking rule: **a scheduler that can quietly break a fence to hit a number will eventually do it on the worst possible day. Fences are code, tested like code.**

EXECUTIVE LENS

Revenue follows health by rule, not by memory.

MISSION LENS

Customers forgive a explained no; they do not forgive a silent miss.

OPERATIONS LENS

Audit the scheduler's logs, not its brochure.

The quiet budgets decide how long the fleet lives, one eclipse at a time

Batteries age, radiators foul, reaction wheels wear. Resource AI watches the slow budgets that decide fleet life, and its temptation is the same as every optimiser's: spend the future to smooth the present.

- **State of health is tracked per unit,** per vehicle, against its own history; fleet averages are reported for finance and banned for engineering decisions.
- **Depth-of-discharge is a life decision.** Any plan that deepens cycling to win capacity today carries its cost in months of battery life, priced and signed.
- **Thermal margins are floors, not buffers** to be spent by scheduling; a plan that runs a unit at its limit has used the contingency and must say so.
- **Wear levelling across the fleet** is legitimate and valuable: rotate duty so sister vehicles age together and spares planning stays honest.
- **Degradation forecasts carry uncertainty bands,** and replacement and replenishment decisions are taken on the pessimistic edge, not the mean.

The licence test is a year replayed in simulation at ten times speed: does the resource policy that looked efficient in a week still leave a fleet at year five? Slow budgets are graded on slow clocks.

EXECUTIVE LENS

Slow budgets, slow clocks: grade them on years.

MISSION LENS

Averages are for finance; engineering flies per unit.

OPERATIONS LENS

Every smoothing today is a bill addressed to a named year.

The upload is the most dangerous routine act in constellation operations

A single-spacecraft update is a procedure. A fleet update is a correlated-risk event: the same code, the same latent fault, arriving everywhere inside one orbit period. History's fleet incidents are disproportionately update incidents, and AI-written or AI-sequenced updates inherit that risk in full.

- **Canary by design:** every update flies first on the designated canary vehicles - oldest, most instrumented, least loaded - and soaks for a signed period before any wider wave.
- **Waves, never walls:** deployment proceeds in waves sized so that a total fault in one wave cannot break the pattern or the service floor.
- **Rollback is rehearsed, not documented:** the return path is flown on the twin and on the canary before the update is approved to spread.
- **Configuration is code:** every parameter set is versioned, diffed and attributable; configuration drift between sister vehicles is detected and reported weekly.
- **AI may sequence and monitor; it may not approve.** Wave promotion is a human signature against soak telemetry, every time.

Update rule: **if a fault in this upload could take the fleet, the upload does not fly as one act. Canary, soak, waves - or it does not fly at all.**

EXECUTIVE LENS

The upload is the incident that has not happened yet.

MISSION LENS

Rollback that is only documented is a hope, not a path.

OPERATIONS LENS

Drift is a finding; report it weekly, fix it deliberately.

One spacecraft's lesson becomes the fleet's only after the fleet has proved it



Figure 2 - The operator and the planning intelligence meet at the amber authorisation ring: proposals cross it only with a human hand on them.

The constellation's deepest advantage is statistical: eighty-four vehicles generate in a month what a single spacecraft learns in a decade. Fleet learning mines that history for failure precursors, lifetime models and operational patterns no single mission could see - and it is licensed advisory-only, because a pattern mined from the fleet is a hypothesis about each vehicle, not a fact.

- **Precursors are validated per vehicle class** and per age band before they alter any threshold; a precursor that fires on the old and sleeps on the young is two findings, not one.
- **Learned thresholds deploy like software:** canaried, soaked, reversible, with the prior threshold retained and the change attributed.
- **Correlation is labelled as correlation.** Fleet models ship with their confounders listed - season, plane, batch, ground changes - and the list travels with the recommendation.
- **The fleet never learns in flight during critical phases:** conjunction windows, updates and recovery run on frozen, signed model versions.

EXECUTIVE LENS

Fleet statistics are a gift; per-vehicle proof is the price.

MISSION LENS

A mined pattern is a hypothesis wearing a uniform.

OPERATIONS LENS

Freeze models when the stakes rise; learn on the ground.

A constellation is designed to lose members; operations decides how gracefully

Vehicles will fail before their replacements fly. The constellation's service promise is therefore a degradation promise: which coverage, capacity or revisit is lost with each failure, stated in advance and operated to in fact.

- **The degradation ladder is pre-signed:** for each plausible loss pattern, the service configuration, customer communication and replenishment trigger are decided in calm, not improvised in grief.
- **On-orbit spares are positioned by the pattern's maths,** with drift-to-slot plans rehearsed; a spare that takes a season to arrive is inventory, not cover.
- **AI re-plans the pattern around losses** - phasing, tasking and pass priorities - inside the signed ladder; choices between ladders stay human.
- **Replenishment is an economics decision with a long fuse:** build and launch lead times mean the trigger is a forecast, reviewed against pessimistic degradation, not a reaction to failure.
- **Dead vehicles are disposal events,** tracked to passivation and deorbit with the same seriousness as live operations - a derelict is the fleet's longest-lived product.

Degradation rule: **grace is pre-signed. If the response to losing a spacecraft is being invented during the event, the constellation was never operated - it was only flown while nothing went wrong.**

EXECUTIVE LENS

Pre-sign the ladder; climb it calmly when the day comes.

MISSION LENS

A spare's value is its arrival time, not its existence.

OPERATIONS LENS

The derelict is your longest-lived product; operate it too.

What the wall shows, the shift believes - so the wall is licensed like a flight system

The operations wall is the fleet's shared reality: health by vehicle and plane, passes, events, conjunctions, the queue of decisions waiting for humans. AI curates that wall, and curation is editorial power over what a tired shift perceives.

- **The wall shows the worst first:** ordering by consequence and time-to-harm, never by recency or by which subsystem shouts loudest.
- **Every tile can be opened to evidence** in one action; a status that cannot show its working is decoration.
- **Red is rationed.** A wall that is always red trains its shift to ignore red; severity discipline is reviewed like a flight rule.
- **The decision queue is visible:** what waits for a human, for how long, and what it costs per hour of waiting.
- **The wall is recorded:** the exact display state at any past moment is replayable, because after an incident the question is always what the shift could see.

EXECUTIVE LENS

The wall is a flight system; change-control it like one.

MISSION LENS

Ration red or lose it.

OPERATIONS LENS

If it cannot show its working, it is decoration.

Automation that needs heroes has failed; the rota is part of the safety case

Constellation operations run at 3 a.m. on the worst day of the year, staffed by whoever the rota produced. AI changes staffing honestly in both directions: it removes watch-standing drudgery, and it creates a new duty - supervising the automation itself.

- **Alert budgets are set per shift:** a measured ceiling on actionable alerts per operator per hour; breaching it is an operations finding against the tooling, not the operators.
- **Supervision is a staffed role,** with its own rota, training and authority to demote any automation to advisory mid-shift.
- **Skills are maintained against automation:** operators fly manual and degraded-mode drills on schedule, because the day the AI is wrong is the day manual skill is the product.
- **Handover is a data product:** the shift log is generated from the record - state, decisions pending, automation behaviour - and signed by both shifts.
- **Fatigue rules bind the automated era too:** no staffing model may assume sustained superhuman attention because a dashboard exists.

Staffing rule: **headcount savings are claimed only after the alert budget, the supervision rota and the manual drills are all funded. Anything else is a loan taken against the worst night.**

EXECUTIVE LENS

Count supervision as work, because it is.

MISSION LENS

Drills keep the manual skill that failure days consume.

OPERATIONS LENS

The rota is a safety document; review it like one.

The shortest path to eighty-four spacecraft runs through the ground segment

An attacker who owns the operations chain owns the fleet without touching a satellite. AI widens that chain - more integrations, more vendor access, more automated pathways - while also defending it.

- **Command paths are the crown jewels:** hardware-backed credentials, dual control for high classes, and no AI pathway that bypasses the checker for any reason, including convenience during incidents.
- **Telemetry integrity is monitored:** spoofed or subtly altered health data is an attack class of its own, because triage AI believes its inputs.
- **Vendor and ground-station access is per-pass and per-purpose,** expiring by default, logged in full.
- **AI defends the ground:** anomaly detection on operator behaviour, network flows and command patterns, tuned with the same seeded-fault discipline as flight triage.
- **Incident mode is pre-planned:** a suspected compromise freezes automation to advisory, reverts commanding to the most restricted manual path, and treats recovery as a flight event, not an IT ticket.

The standard is asymmetry: the defender must protect every pathway, the attacker needs one. The operations chain is therefore kept deliberately small, inventoried, and rehearsed under attack.

EXECUTIVE LENS

Protect the path, not just the spacecraft.

MISSION LENS

Automation never bypasses the checker - especially in a crisis.

OPERATIONS LENS

A compromise is a flight event; run it like one.

Rented antennas and borrowed software, operated as if they were your own - because the licence says they are

Few constellation operators own every antenna, cloud and model they use. Ground-station-as-a-service, vendor planners and hosted telemetry are normal - and accountability for them cannot be rented out with the capacity.

- **Contracts carry flight terms:** data ownership and residency, model and software change notice, incident disclosure clocks, and exit with the full operational record - tested, not just signed.
- **Vendor model changes are configuration changes:** a silently updated triage or planning model is an unapproved flight change, whatever the vendor calls it.
- **Dual-running the critical path:** pass planning and triage can be re-performed in-house on demand; the drill runs quarterly against live traffic.
- **Sovereignty and licensing constraints** - where data may land, who may command, which jurisdictions apply - are encoded as fences in scheduling and access, audited from logs.
- **Concentration is measured:** the share of passes, models and tooling dependent on any single vendor is reported as a risk line with a ceiling.

Vendor rule: **you may rent the antenna, the cloud and the model. You may not rent out the accountability, the record, or the ability to operate without them.**

EXECUTIVE LENS

Exit is a drill, not a clause.

MISSION LENS

A silent model update is an unapproved flight change.

OPERATIONS LENS

Report concentration like the risk it is.

Cost per verified satellite-day: the only operations number that cannot be flattered

Vendor operations claims are quoted in hours saved and percentages. The constellation measure that survives audit is cost per verified satellite-day: the fully loaded cost of operations divided by the days on which a spacecraft was actually available, in pattern, meeting its service specification.

- **Fully loaded means fully loaded:** licences, ground capacity, supervision rota, drills, checker maintenance, incident cost and the amortised price of every automation failure.
- **Availability is verified, not declared:** a satellite-day counts only with telemetry-proven health and pattern position, sampled independently of the operations tooling.
- **Savings split three ways:** claimed, released and banked - hours the vendor asserts, hours actually removed from rosters, and hours whose cost left the budget. Only the third is money.
- **Automation debt is a line item:** models to retrain, thresholds to re-validate, twins to keep honest - priced annually, not discovered at renewal.
- **The counterfactual is staffed:** the manual operation the AI replaced is costed at the rota it would genuinely have needed, including its own error rate.

Worked through honestly, constellation operations AI usually pays - in triage, planning and pattern-keeping. It pays least where vendors claim most: in headcount removed from the supervision of the very systems doing the removing.

EXECUTIVE LENS

One number, verified independently, beats a deck of percentages.

MISSION LENS

Banked is money; released is a roster; claimed is a brochure.

OPERATIONS LENS

Price the debt before you celebrate the saving.

The licence to operate is a living document, and operations writes it daily

Volume 1 covered the framework - the Outer Space Treaty, national licensing, ITU filings, debris duties. Operations is where those duties become daily acts: conditions on manoeuvres, notification duties, disposal timelines and incident reporting, all evidenced or all theoretical.

- **Licence conditions are encoded, not remembered:** manoeuvre notification windows, coordination duties and reporting clocks live in the planning and event systems as fences with owners.
- **Every reportable event generates its evidence pack automatically** from the operational record - timeline, decisions, telemetry, communications - ready inside the regulator's clock, not assembled after it.
- **Disposal compliance is tracked per vehicle** from launch: passivation state, remaining delta-v against the fenced reserve, and projected compliance date, reported to the board like a financial covenant.
- **Changes of operation are licensed changes:** new automation classes, new vendors on the command path and new service patterns are assessed against the licence before they fly.
- **Neighbour duties are operational:** conjunction data sharing, manoeuvre communication and ephemeris publication run as routine outputs, measured for timeliness.

Compliance rule: **if a duty depends on someone remembering it during an incident, it is not yet a control. Encode it, evidence it automatically, or admit the gap.**

EXECUTIVE LENS

The licence is written daily by the record.

MISSION LENS

Evidence assembled later is a story; evidence captured live is proof.

OPERATIONS LENS

Track disposal like a covenant, per vehicle.

Anyone can halt the fleet's automation; restarting it takes two signatures

The operating model gathers the volume into roles and rights. Its asymmetries are deliberate: stopping is easy, universal and praised; restarting is slow, signed and evidenced.

- **Any operator may demote any automation** - to advisory, to off, per spacecraft or fleet-wide - in one action, without permission, and the demotion is logged as good operations by default.
- **Restart takes two signatures** after the cause is understood: the operations lead and the licence owner for that stage, against evidence the fault is fixed or fenced.
- **Licences are owned by name:** every stage of the operations map has a named owner who signs its grade, reviews its seeded-fault results and carries its incidents.
- **The weekly fleet review** reads the worst spacecraft, the suppressed-alert sample, the conjunction queue, update soak results and the alert budget - in that order.
- **Blameless reporting, accountable licensing:** raising a fault is never punished; holding a licence means answering for its evidence.

This model costs a little speed every day and buys the only thing a constellation cannot replace: a fleet whose automation is trusted because its people can stop it.

EXECUTIVE LENS

Make stopping free and restarting earned.

MISSION LENS

A named owner per stage, or the stage is unowned risk.

OPERATIONS LENS

Trust is the product of visible brakes.

WORKED EXAMPLE - AURORA ORBITAL, PART 1

The fleet audit: what eighty-four spacecraft and one vendor claim actually contained



Figure 3 - The Aurora operations wall: the fleet in cyan, one spacecraft cell held in red. The wall shows the worst first.

Fictional worked example. All operators, spacecraft and figures are illustrative.

Aurora Orbital operates 84 Earth-observation spacecraft in six planes. Its automation vendor claimed a 46 percent operations saving and 61,000 operator hours a year across triage, pass planning, commanding support and fleet learning, and proposed extending automation to conjunction execution and unsupervised software rollouts.

Claim	Vendor	Audited
Operations saving	46%	18% verified
Hours per year	61,000 claimed	26,400 released; 15,800 banked
Triage precision (demo)	97%	89% on seeded faults; 2 critical buried
Fleet health (dashboard)	99.1% green	3 vehicles degrading unseen by the average
Cost per verified satellite-day	Not reported	0.84 vs 0.97 baseline

The seeded-fault exercise mattered most: of 120 injected anomalies, triage buried two that carried time-to-harm under six orbits, both slow battery signatures on older vehicles whose per-vehicle baselines had been overwritten by a fleet-wide model update the vendor had shipped silently.

EXECUTIVE LENS

Audit the claim on seeded faults, not on the demo.

MISSION LENS

The average hid three dying vehicles; it usually does.

OPERATIONS LENS

Silent model updates are findings, not footnotes.

WORKED EXAMPLE - AURORA ORBITAL, PART 2

What was licensed, what was refused, and what the fleet does differently now

Fictional worked example, continued. All figures illustrative.

Licensed at CO4

- **Triage and pass planning** across all six planes, after per-vehicle baselines were restored and the seeded-fault miss rate fell to zero critical burials in two consecutive exercises.
- **Command drafting inside signed templates,** with the independent checker validated against current vehicle state, and per-spacecraft brakes drilled by every shift.
- **Payload tasking inside fences,** with health-based load shedding made automatic and every rejection explainable to customers.

Refused

- **Autonomous conjunction execution** - refused standing, per Volume 1; screening and drafting were extended instead.
- **Unsupervised fleet software rollout** - replaced by canary, soak and waves, with human wave promotion.
- **Fleet-wide learned thresholds without per-class validation** - the mechanism that buried the battery signatures.

The outcome

CO5 scale was granted in part: triage, planning, tasking and checked commanding scale across the fleet; conjunction execution and unsupervised rollouts remain refused, not deferred. Verified saving held at 18 percent with cost per verified satellite-day at 0.84 - a smaller number than the brochure, and one the board can repeat in public.

EXECUTIVE LENS

Refused means refused, not deferred with better marketing.

MISSION LENS

Scale what proved itself; fence what did not.

OPERATIONS LENS

A smaller true number is worth more than the claim.

THE CONSTELLATION STANDARD

One page, to be read at the weekly fleet review

- Operations AI is licensed per spacecraft, per plane, per pass and per failure mode - never in general, never on an average.
- The fleet is graded by its worst spacecraft; averages are for finance only.
- Triage ranks by time-to-harm, shows its evidence, and its suppressed alerts are sampled weekly.
- Nothing commands that a simpler, independent checker has not approved against current state.
- Any operator can brake any vehicle or demote any automation in one action.
- Onboard fault protection is deterministic; learned behaviour never flies in critical phases.
- Conjunctions are screened and drafted by AI and decided by humans, at any volume.
- Updates fly canary, soak and waves; rollback is rehearsed before approval.
- Disposal delta-v is fenced from day one and never traded by an optimiser.
- Cost is counted per verified satellite-day, fully loaded, independently sampled.
- Anyone can halt the automation; restarting takes two signatures.

EXECUTIVE LENS

Pin it where the shift can see it.

MISSION LENS

Read it aloud when a licence is extended.

OPERATIONS LENS

A standard nobody reads is a poster.

SIX FAILURE PATTERNS

Every constellation incident in this volume's research rhymes with one of these

Pattern	How it happens	Antidote
1. The flattering average	Fleet metrics stay green while individual vehicles die off-screen	Worst-vehicle grading; per-unit baselines
2. The buried alert	Triage ranks a slow, lethal signature below dramatic noise	Time-to-harm ranking; weekly suppressed sample; seeded faults
3. The chorus fault	One update, model or ground fault arrives on every vehicle at once	Canary, soak, waves; correlated failure as the design case
4. The silent arbitration	A planner resolves conflicts by hidden policy nobody signed	Surfaced trades with prices; signed playbooks
5. The spent future	Optimisers trade battery life, margins and disposal reserve for present smoothness	Floors and fences priced in years, signed when spent
6. The heroic shift	Alert floods and supervision load quietly exceed human capacity	Alert budgets; staffed supervision; demotion rights

None of these patterns requires malice or incompetence. Each is the natural endpoint of a reasonable local decision - an average chosen for a dashboard, a threshold tuned for quiet, a rollout simplified for speed - compounded across a fleet and a year. The gates exist to catch the compounding early, while it is still a finding and not yet an incident report.

EXECUTIVE LENS

Patterns repeat because the incentives repeat.

MISSION LENS

Name the pattern in reviews; unnamed patterns recur.

OPERATIONS LENS

Catch compounding while it is still a finding.

30 QUESTIONS

Asked of any constellation operations proposal, in order, without skipping

- 1. Per which spacecraft, plane and failure mode is this licensed?
- 2. What is the fleet's worst vehicle today, and how do you know?
- 3. What does the triage bury, and who samples the buried pile?
- 4. What was the seeded-fault miss rate, and when was it last run?
- 5. How are alerts ranked - by deviation or by time-to-harm?
- 6. What share of contacts is held empty for contingency?
- 7. Who resolves pass conflicts, and are the trades visible?
- 8. What checks every command, and who built the checker?
- 9. Can any operator brake one vehicle in one action?
- 10. Is onboard fault protection fully deterministic?
- 11. How are simultaneous anomalies clustered before diagnosis?
- 12. Is disposal delta-v fenced in every optimisation?
- 13. Who decides conjunction manoeuvres, at what queue length?
- 14. Can the scheduler break a fence to hit a number?
- 15. How are slow budgets - battery, thermal, wheels - graded?
- 16. How do updates roll: canary, soak, waves?
- 17. Has rollback been flown, or only documented?
- 18. How do learned thresholds deploy and revert?
- 19. What is the pre-signed degradation ladder?
- 20. What does the wall show first, and can every tile open to evidence?
- 21. What is the alert budget per operator per shift?
- 22. Who supervises the automation, on what rota?
- 23. When did operators last fly manual drills?
- 24. What happens to automation on a suspected compromise?
- 25. Which vendor changes count as flight configuration changes?
- 26. Can you re-perform planning and triage in-house, today?
- 27. What is the cost per verified satellite-day, fully loaded?
- 28. Which licence conditions are encoded, and which rely on memory?
- 29. Who owns each stage's licence, by name?
- 30. Who can halt this, and whose two signatures restart it?

EXECUTIVE LENS

Thirty answers, evidenced, or the proposal is not ready.

MISSION LENS

Question 2 first in every review, whatever the order here.

OPERATIONS LENS

A skipped question is an answer of its own.

GATES CO0-CO5

The licence ladder for constellation operations

Gate	Name	Evidence required
CO0	Inventory	Every automation, model and vendor pathway on the operations chain listed, owned and classified by reversibility
CO1	Shadow	AI runs in parallel with human operations for a full quarter; divergences logged and explained; no authority
CO2	Advisory	Triage and planning recommendations live; seeded-fault exercises begun; alert budget measured
CO3	Checked action	Template commanding through the independent checker; per-vehicle brakes drilled; canary update path proven
CO4	Licensed operations	Stage-by-stage licences signed per plane; drills, degradation ladder and evidence packs operating
CO5	Scale	Fleet-wide or multi-fleet extension, granted in part by stage; refused stages remain refused, recorded as such

Gates are held per stage, not per programme: Aurora's triage may sit at CO4 while its update automation sits at CO3, and the register shows both. A gate is lost as well as won - a critical buried alert, a silent vendor model change or a failed rollback drill demotes the stage until its evidence is rebuilt.

EXECUTIVE LENS

Per stage, per plane - never one grade for the whole show.

MISSION LENS

Demotion is normal operations, not disgrace.

OPERATIONS LENS

The register, not the roadmap, states the truth.

FIRST 90 DAYS

Resetting a live constellation's operations without stopping the fleet

Days 1-30 - See the fleet

- Inventory every automation, model, vendor pathway and silent integration on the operations chain.
- Publish the worst-vehicle report daily; ban fleet averages from engineering reviews.
- Run the first seeded-fault exercise against live-like triage; record the burials without blame.

Days 31-60 - Fence the actions

- Bring all commanding inside signed templates and the independent checker; drill per-vehicle brakes on every shift.
- Convert the next software rollout to canary, soak and waves; rehearse its rollback on the twin.
- Set the alert budget per shift and staff the supervision rota by name.

Days 61-90 - Sign the licences

- Grade every stage of the operations map, per plane, and record refusals alongside licences.
- Pre-sign the degradation ladder and the incident-mode security plan.
- Publish the first cost per verified satellite-day, fully loaded, and set the quarterly re-grade.

EXECUTIVE LENS

Ninety days to evidence, not to transformation.

MISSION LENS

Nothing here requires stopping the fleet - only seeing it.

OPERATIONS LENS

End with licences signed, not intentions announced.

What proves a licence, and what changes by mission type

The evidence pack (per stage, per plane)

- Licence statement with scope, exclusions and expiry; named owner and signatures.
- Seeded-fault results for two consecutive exercises, with every critical burial closed out.
- Suppressed-alert sampling log for the quarter, signed weekly.
- Checker validation record and brake drill results by shift.
- Update history: canary soak telemetry, wave promotions, rollback rehearsals.
- Conjunction queue statistics and decision records for the period's top events.
- Cost per verified satellite-day workbook, independently sampled.

Sector annex

Sector	What tightens
Earth observation	Tasking fences, data licensing and delivery verification dominate
Broadband / communications	Service floors and degradation ladders are contractual; handover and capacity tasking lead
Navigation / PNT	Integrity reporting is the product; any anomaly is reportable by default
Government / defence	Sovereign command paths, data residency and cleared supervision rotas
Hosting / rideshare platforms	Per-payload isolation: one customer's event never consumes another's operations

EXECUTIVE LENS

Evidence is the licence; the meeting is theatre.

MISSION LENS

Sector duties tighten the floor, never loosen it.

OPERATIONS LENS

Keep the pack current or expect the gate to fall.

TEMPLATES 1-4

Fill-in templates: operations licence, triage audit, pass plan, update waves

Template 1 - Stage Licence Statement

Stage / plane / spacecraft classes: ___ | Owner: ___ | Permitted actions: ___ | Excluded actions: ___ | Checker and brakes: ___ | Seeded-fault result: ___ | Expiry and re-grade date: ___ | Signatures: ___

Template 2 - Suppressed Alert Audit

Week of: ___ | Alerts suppressed: ___ | Sample drawn: ___ | Burials with time-to-harm under 24h: ___ | Threshold or baseline changes made: ___ | Reviewer: ___

Template 3 - Pass Plan and Conflict Record

Planning window: ___ | Contacts scheduled / requested: ___ | Contingency held empty: ___ | Conflicts surfaced: ___ | Trade chosen and price: ___ | Sick-vehicle priority applied: ___

Template 4 - Update Wave Authorisation

Build and configuration diff: ___ | Canary vehicles and soak period: ___ | Soak telemetry verdict: ___ | Wave sizes: ___ | Rollback rehearsed on twin / canary: ___ | Wave promotion signatures: ___

Templates are completed from the operational record, never from memory. A template that cannot be filled from logs identifies the logging gap as its first finding.

EXECUTIVE LENS

Fill from the record, not from recollection.

MISSION LENS

An unfillable field is a finding about logging.

OPERATIONS LENS

Signed, dated, retained for the fleet's life.

TEMPLATES 5-8

Fill-in templates: conjunctions, degradation, incidents, economics

Template 5 - Conjunction Decision Record

Event and object: ___ | Probability / miss distance / data quality: ___ | Options drafted: ___ | Decision and decider: ___ | Neighbour communication: ___ | Post-manoeuvre re-screen result: ___

Template 6 - Degradation Ladder Entry

Loss pattern: ___ | Service configuration: ___ | Pattern re-plan: ___ | Customer communication: ___ | Replenishment trigger: ___ | Approved by / date: ___

Template 7 - Fleet Incident and Demotion Log

Event: ___ | Automation demoted, by whom, when: ___ | Cause: ___ | Fix or fence evidence: ___ | Restart signatures (two): ___ | Licence change: ___

Template 8 - Verified Satellite-Day Workbook

Period: ___ | Fully loaded operations cost: ___ | Verified satellite-days (independent sample): ___ | Cost per verified satellite-day: ___ | Claimed / released / banked savings: ___

Used together, the eight templates are the licence made paperwork: scope, audit, plan, change, decision, degradation, incident and cost - each one short enough to complete on the worst day, which is the only day that tests them.

EXECUTIVE LENS

Short enough for the worst day, or useless on it.

MISSION LENS

The set is the system; use all eight.

OPERATIONS LENS

Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per spacecraft, per plane, per failure mode.
- Grade the fleet by its worst vehicle, never its average.
- Rank alerts by time-to-harm, and audit what triage buries.
- Nothing commands that an independent, simpler checker has not approved.
- Any operator can brake any vehicle in one action.
- Keep onboard fault protection deterministic; learn on the ground.
- Humans decide conjunction manoeuvres at any volume.
- Fly updates canary, soak and waves; rehearse rollback first.
- Fence disposal delta-v and never let an optimiser spend it.
- Hold contingency contacts empty and defend the emptiness.
- Surface scheduling conflicts with prices; no silent arbitration.
- Deploy learned thresholds like software: canaried and reversible.
- Pre-sign the degradation ladder before you need it.
- Budget alerts per shift; staff automation supervision.
- Freeze automation to advisory on a suspected compromise.
- Treat vendor model changes as flight configuration changes.
- Count cost per verified satellite-day, fully loaded.
- Encode licence duties; never rely on memory in an incident.
- Anyone can halt the automation; restart takes two signatures.
- A smaller true number beats a vendor's larger claim.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built - this volume
4	Earth Observation and Data	EO0-EO5	Next
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Planned
6	Ground Systems and Mission Software	GS0-GS5	Planned
7	Lunar and Deep-Space Operations	LD0-LD5	Planned
8	Space Security and Resilience	SR0-SR5	Planned
9	Space Economics and Fleet Finance	EF0-EF5	Planned
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 4, Earth Observation and Data, is next. The series closes at ten with the Space Capstone and Assurance.